

***In vitro* testing of guide RNA efficiency for CRISPR-mediated genome editing**

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In CRISPR-mediated genome editing, guide RNAs (gRNAs) serve to target the Cas9 nuclease to an intended target genomic locus. Many gRNAs are not efficient in facilitating Cas9-mediated cleavage by introduction of double-strand breaks. Therefore, after designing possible gRNA sequences via web-based tools, it is critical to assess their cleavage efficiency at the target locus and identify the most efficient gRNA(s) *in vitro* before deploying in binary plasmids for genome editing *in planta*.

Reagents necessary:

CTAB extraction buffer for genomic DNA extraction
Nuclease free water
Equipment and reagents to run DNA products through standard agarose gel electrophoresis
Phusion DNA polymerase kit or any high-fidelity DNA polymerase
HiScribe T7 High Yield RNA Synthesis Kit (NEB E2050S)
Cas9 nuclease (NEB Cas9 M0386S)

Method:

1. Cloning a template fragment from genomic DNA

- a) Perform DNA extraction using CTAB method or any other methods that you may use routinely for your plant species.
- b) Design primers flanking your target PAM sites (NGG). PCR to amplify around 500-600 bp of genomic/gene fragment around the target. Large size templates are easy to resolve on gel electrophoresis.
- c) Clean PCR product using Zymo/Bioneer DNA cleanup kit and Sanger sequence to confirm that your PCR amplifications are intended gene. This will also help later for screening indel polymorphisms at PAM site.

2. Designing gRNA and synthesis of guide RNA using *in vitro* T7 kit

Design targets for editing target exon(s) using gRNA design tool (<http://crispr.mit.edu/> or any other portal). The following primers are then used to synthesize the DNA template to prepare gRNA for each target.

Oligo design

- a. Primer 1 (specific for each guide):
 - a) TAATACGACTCACTATAG**GGG**-(20 bp guide sequence)-gtttAagagctaTGCTGgaa
 - b) The three bolded G's are included in the transcript and required for transcription. If the guide-specific sequence starts with one or more G's, the same number of bolded G's can be deleted from the primer.
- b. Primer 2 (always the same):
 - a) Reverse-gRNA Template:
aaaagcaccgactcgggtgccacttttcaagttgataacggactagccttatttAaacttgctaTGCTGttt
cCAGCAtagctctTaaac

PCR Double-stranded DNA synthesis for gRNA

- a. Set up the following PCR reaction
 - 30.5 uL nuclease free water
 - 10 uL 5X Phusion buffer
 - 1 uL 10 mM dNTP
 - 4 uL primer 1 (5 uM)
 - 4 uL primer 2 (5 uM) Always Primer 2 (Reverse-gRNA Template)

0.5 uL Phusion polymerase

Total 50 uL

- b. Program for PCR
98 °C 1 min

98 °C 30s

55 °C 15s 8X-10X cycles of these three steps

72°C 1 min

72 °C 5 min

4 °C hold

Nano-drop to quantify the PCR product. Run a small aliquot of the PCR product out on a gel to check the bands. The resulting band should be 90bp (very small) and so it is necessary to run a ≥3% agarose gel.

- c) Purify the PCR product using Zymo/bioneer DNA cleanup kit.

3. gRNA synthesis

Use HiScribe T7 High Yield RNA Synthesis Kit (NEB E2050S)

- a. Set up reaction in PCR tube
7 uL water (nuclease free)
5 uL NTP mix 10 mM each
2 uL Template DNA (~100 to 500 ng)
1 uL T7 RNA Polymerase Mix

Total 15 uL

- b. Incubate at 37 °C for 2 hours
c. Move the reaction into 1.5mL eppendorf tube
d. Add 435 uL DNAase/RNase free water
e. Measure gRNA concentration on nanodrop. Note it down. (May not be accurate, as NTPs may inflate the nanodrop readings).
f. Optional to cleanup using any RNA clean-up kit. Dilute to 10 ng / uL by adding nuclease free water.

4. Cas9 nuclease assay and gel electrophoresis

Incubation of template DNA with gRNA and Cas9 nuclease (NEB Cas9 M0386S)

- a. Set up reaction (include a negative RNA control)
7.5 uL water (nuclease free)
1.5 uL 10X Cas9 reaction buffer
1.5 uL 300 nM sgRNA (10 ng / uL)
0.5 uL Cas9 Nuclease

Total 11uL

- b. Pre-incubate for 10 min at 37 °C.
c. Add 4 uL substrate DNA (~50-100 ng total)
d. Incubate 1 hr at 37 °C (or longer up to 1.5 hr)
e. Denature Cas9 by incubation at 80°C for 10 min

Set up a control reaction including all the above ingredients without gRNA. In addition, different ratios of gRNA to Cas9 nuclease can also be setup to replicate the efficiency tests.

Gel Electrophoresis

Prepare 1.1% agarose gel. Mix the Cas9 incubated reaction with 2uL of loading dye. Load the mixture on gel. If stain was not added during gel preparation, soak gel in staining buffer. Run gel electrophoresis until the ladder resolves and check the cleavage of template DNA under UV illumination.